

One implication of this Gamma convergence is that: if u_n minimizes $\tilde{\mathcal{E}}_n$ over $g + W^{1,n}(\Omega)$, then any cluster point, say u , of u_n should minimize $\tilde{\mathcal{E}}_\infty$, i.e.

$$u = \arg \min_{w \in g + W_0^{1,\infty}(\Omega)} \|\nabla w\|_{L^\infty}.$$

Hence u is infinity harmonic, i.e. the Gamma convergence of $\tilde{\mathcal{E}}_n$ echoes the convergence of u_n . This can be regarded as an alternative proof for the classical result that $-\Delta_p$ is approximately an infinity Laplacian when the constant exponent p is large.

Coming back to the Orlicz setting, in general, we cannot simply replace $\|\nabla u\|_{L^n}$ with $\|\nabla u\|_{L^{\Phi_n(\cdot)}}$ and expect a clean analogy for the above to hold. One reason is that the Luxemburg norm of an Orlicz space is not a complete parallel for the usual L^p space. For example, $\Phi(x, s) = s^p$ ($p \geq 1$) a generalized Φ -function, its Luxemburg norm $\|\cdot\|_{L^{\Phi(\cdot)}}$ is equivalent but not equal to $\|\cdot\|_{L^p}$. Some scalings are needed to make them agree. For detailed discussions of the scalings of generalized Orlicz spaces and its consequences on Luxemburg norms, we refer to [HH19, Section 3.4]. That being said, there are some situations where such analogy exists.

Theorem 11. *Assume that Φ_n satisfies (4.3) and the unit normalization condition (6.4). Assume in addition that (a stronger version of (4.4) holds, i.e.)*

$$\lim_{n \rightarrow \infty} p_n^- = \infty \text{ and } \lim_{n \rightarrow \infty} \frac{p_n^+}{p_n} = 1.$$

Let

$$\tilde{E}_n : L^1(\Omega) \mapsto \begin{cases} \|\nabla u\|_{L^{\Phi_n(\cdot)}}, & u \in W^{1,\Phi_n(\cdot)}(\Omega); \\ \infty, & \text{otherwise,} \end{cases}$$

and

$$\tilde{E}_\infty : L^1(\Omega) \mapsto \begin{cases} \|\nabla u\|_{L^\infty}, & u \in W^{1,\infty}(\Omega); \\ \infty, & \text{otherwise.} \end{cases}$$

Then $\Gamma(L^1(\Omega))\text{-}\lim_{n \rightarrow \infty} \tilde{E}_n = \tilde{E}_\infty$.

This can be proved by similar arguments as [BM15, Theorem 2]. We omit the details.

7. COMPARISON PRINCIPLE AND UNIQUENESS OF THE LIMIT EQUATION

Let u_n be the sequence of the unique continuous weak solution for equation (5.3). In Theorem 8 or Theorem 10, we have established that $\{\|\nabla u_n\|_{L^m} : p_n^- \geq m\}$ is bounded for all $m > d$ and there is a subsequence of u_n which converges uniformly to the following limit equation

$$(7.1) \quad \begin{cases} -\Delta_{\infty,\Lambda} u = 0, & x \in \Omega; \\ u = g, & x \in \partial\Omega, \end{cases}$$

where

$$-\Delta_{\infty,\Lambda} u = -\Delta_\infty u - |\nabla u| \langle \Lambda(x, |\nabla u|), \nabla u \rangle.$$

Remark 7. *We shall show that the equation (7.1) has a unique continuous viscosity solution. Since u_n is continuous and $\{\|\nabla u_n\|_{L^m} : p_n^- \geq m\}$ is bounded for all $m > d$, the uniform convergence limit of every subsequence of u_n is again continuous. Hence the uniqueness result indicates that the whole sequence u_n , not just a subsequence, converges uniformly to the unique continuous viscosity solution of (7.1).*